

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0044

Poages Mill 138 kV
October 16, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0044. The Interconnection Customer proposes a 491.3 MW solar photovoltaic facility interconnecting at the Poages Mill 138 kV substation in the Virginia area. The request was received on June 13, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 39 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$94,182,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Floyds Knobs - Poages Mill 138kV
Contingency: Leesville - Reusens 345kV
Loading: 237.1% (Rating: 223 MVA)
DFAX: 72.0%
Pre-project loading: 78.5%
2. Facility: Meadowview - Pages Mill 138kV line
Contingency: Poages Mill - Floyds Knobs 138kV
Loading: 206.9% (Rating: 232 MVA)
DFAX: 100.0%
Pre-project loading: 5.0%
3. Facility: Hancock - Floyds Knobs 138kV
Contingency: Reenergy Livermore Falls - Floyds Knobs 138kV
Loading: 195.4% (Rating: 213 MVA)
DFAX: 74.5%
Pre-project loading: 23.2%
4. Facility: Rally Hill - Meadowview 138kV L
Contingency: Poaes Mill - Floyds Knobs 138kV
Loading: 183.6% (Rating: 246 MVA)
DFAX: 100.0%
Pre-project loading: 16.4%
5. Facility: Train - Rally Hill Substation 138kV Line
Contingency: Floyds Knobs - Poages Mill 138kV line
Loading: 175.2% (Rating: 250 MVA)
DFAX: 100.0%
Pre-project loading: 20.9%

6. Facility: Train - Asheville 138kV line
Contingency: Floyds Knobs - Poages Mill 138kV
Loading: 155.3% (Rating: 273 MVA)
DFAX: 100.0%
Pre-project loading: 24.6%
7. Facility: College - Poplar-Antelope 138kV
Contingency: Floyds Knobs - Hancock 138kV
Loading: 133.4% (Rating: 160 MVA)
DFAX: 40.6%
Pre-project loading: 8.9%
8. Facility: Ash Street - Hancock 138kV line
Contingency: Floyds Knobs Substation - Reenergy Livermore Falls 138kV
Loading: 124.2% (Rating: 232 MVA)
DFAX: 15.5%
Pre-project loading: 91.4%
9. Facility: Beardsley St - Neosho Substation 138kV
Contingency: Reenergy Livermore Falls - Floyds Knobs 138kV
Loading: 121.0% (Rating: 204 MVA)
DFAX: 15.5%
Pre-project loading: 83.8%
10. Facility: Nacoochee - Boonsboro Substation 138kV
Contingency: Clover - Bear Substation 345kV Line
Loading: 120.7% (Rating: 255 MVA)
DFAX: 8.8%
Pre-project loading: 103.7%
11. Facility: Kumis - College Substation 138kV
Contingency: Hancock - Floyds Knobs 138kV
Loading: 119.2% (Rating: 187 MVA)
DFAX: 40.6%
Pre-project loading: 12.8%
12. Facility: Reenergy Livermore Falls - Kumis 138kV line
Contingency: Hancock - Floyds Knobs 138kV
Loading: 117.2% (Rating: 199 MVA)
DFAX: 40.6%
Pre-project loading: 17.2%
13. Facility: Reenergy Livermore Falls - Floyds Knobs 138kV
Contingency: Floyds Knobs - Hancock 138kV
Loading: 116.6% (Rating: 226 MVA)
DFAX: 40.6%
Pre-project loading: 28.3%
14. Facility: Franklin-Elk Grove - New London 138kV
Contingency: Clover - Bear 345kV Line
Loading: 116.4% (Rating: 222 MVA)
DFAX: 8.9%
Pre-project loading: 96.7%
15. Facility: S Texa - Nacoochee 138kV
Contingency: Clover - Bear 345kV
Loading: 112.6% (Rating: 276 MVA)
DFAX: 8.8%
Pre-project loading: 96.9%
16. Facility: Osceola Substation - Michigan Power Lp 138kV
Contingency: Clover - Bear 345kV

- Loading: 109.1% (Rating: 214 MVA)
DFAX: 9.9%
Pre-project loading: 86.4%
17. Facility: Jack Ramey Wind - Starkey 138kV L
Contingency: Floyds Nobs - Poages Mill 138kV
Loading: 107.8% (Rating: 250 MVA)
DFAX: 24.3%
Pre-project loading: 60.0%
18. Facility: Starkey - Hancock 138kV
Contingency: Floyds Knobs - Poages Mill Substation 138kV Line
Loading: 106.5% (Rating: 227 MVA)
DFAX: 24.3%
Pre-project loading: 53.8%
19. Facility: Safeway Stores - Asheville 138kV
Contingency: Floyds Knobs - Poages Mill 138kV
Loading: 105.5% (Rating: 328 MVA)
DFAX: 24.3%
Pre-project loading: 69.1%
20. Facility: Franklin-Elk Grove - S Texa 138kV
Contingency: Desert Outpost - S Texa 138kV
Loading: 105.5% (Rating: 197 MVA)
DFAX: 7.2%
Pre-project loading: 87.6%
21. Facility: Lone Tree - Osceola 138kV
Contingency: Bear Substation - Clover 345kV
Loading: 105.1% (Rating: 217 MVA)
DFAX: 9.9%
Pre-project loading: 82.8%
22. Facility: N Island - Clearbrook 138kV Line
Contingency: Floyds Knobs - Poages Mill 138kV line
Loading: 104.2% (Rating: 302 MVA)
DFAX: 24.3%
Pre-project loading: 64.6%
23. Facility: Roanoke Electric Steel - Muddy Ranch 138kV
Contingency: Huntington Court 345/138kV TX
Loading: 103.5% (Rating: 239 MVA)
DFAX: 14.1%
Pre-project loading: 74.6%
24. Facility: N Island - Shutesbury 138kV
Contingency: Floyds Knobs - Poages Mill 138kV Line
Loading: 102.8% (Rating: 312 MVA)
DFAX: 24.3%
Pre-project loading: 64.6%
25. Facility: Safeway Stores - Shutesbury 138kV L
Contingency: Poages Mill - Floyds Knobs 138kV L
Loading: 102.4% (Rating: 326 MVA)
DFAX: 24.3%
Pre-project loading: 65.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Poages Mill - Floyds Knobs Substation 138kV	Rebuild line with higher-capacity conductor	\$4,609,000	100% (\$4,609,000)
Meadowview - Poages Mill 138kV	Rebuild line with higher-capacity conductor	\$4,232,000	100% (\$4,232,000)
Hancock - Floyds Knobs 138kV	Rebuild line with higher-capacity conductor	\$3,056,000	100% (\$3,056,000)
Rally Hill - Meadowview 138kV	Rebuild line with higher-capacity conductor	\$2,158,000	100% (\$2,158,000)
Train - Rally Hill Substation 138kV Line	Rebuild line with higher-capacity conductor	\$6,892,000	100% (\$6,892,000)
Train - Asheville 138kV line	Rebuild line with higher-capacity conductor	\$7,967,000	100% (\$7,967,000)
College - Poplar-Antelope 138kV	Rebuild line with higher-capacity conductor	\$7,662,000	100% (\$7,662,000)
Ash Street - Hancock 138kV	Rebuild line with higher-capacity conductor	\$4,022,000	100% (\$4,022,000)
Beardsley St - Neosho Substation 138kV	Rebuild line with higher-capacity conductor	\$4,330,000	100% (\$4,330,000)
Boonsboro Substation - Nacoochee 138kV Line	Rebuild line with higher-capacity conductor	\$9,307,000	100% (\$9,307,000)
Kumis - College Substation 138kV	Rebuild line with higher-capacity conductor	\$7,493,000	100% (\$7,493,000)
Reenergy Livermore Falls - Kumis 138kV line	Rebuild line with higher-capacity conductor	\$6,668,000	100% (\$6,668,000)
Reenergy Livermore Falls - Floyds Knobs 138kV	Rebuild line with higher-capacity conductor	\$2,153,000	100% (\$2,153,000)
Franklin-Elk Grove - New London 138kV	Rebuild line with higher-capacity conductor	\$3,444,000	30% (\$1,019,000)
S Texa - Nacoochee 138kV	Rebuild line with higher-capacity conductor	\$5,362,000	100% (\$5,362,000)
Osceola - Michigan Power Lp 138kV Line	Reconductor line	\$4,459,000	34% (\$1,522,000)
Jack Ramey Wind - Starkey 138kV L	Reconductor line	\$1,048,000	100% (\$1,048,000)
Starkey - Hancock 138kV	Reconductor line	\$791,000	100% (\$791,000)
Safeway Stores - Asheville 138kV	Reconductor line	\$6,281,000	100% (\$6,281,000)
Franklin-Elk Grove - S Texa 138kV	Reconductor line	\$4,047,000	35% (\$1,403,000)
Osceola Substation - Lone Tree Substation 138kV	Reconductor line	\$1,990,000	34% (\$679,000)
N Island - Clearbrook 138kV	Reconductor line	\$2,575,000	100% (\$2,575,000)
Roanoke Electric Steel - Muddy Ranch 138kV	Reconductor line	\$370,000	100% (\$370,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Island - Shutesbury 138kV	Reconductor line	\$1,431,000	100% (\$1,431,000)
Safeway Stores - Shutesbury 138kV L	Reconductor line	\$1,152,000	100% (\$1,152,000)
	Total allocated to project		\$94,182,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.